

LEVELHULL · REQUIREMENTS

LevelHull requirements

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	LVH-REQ-001	Version	0.2
Status	Draft	Date	2026-10-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/levelhull		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-03	Amish Chadha	Targets fixed on the reference canoe; R11 to R13 added; TRL 3 status from LVH-CAL-001
0.1	2026-09-30	Amish Chadha	Initial scaffold

LevelHull requirements

Requirements for the kit fitted to the reference canoe (8.0 m, 1.6 m beam, 0.65 m deep, 15 hp outboard, four crew). Other hulls are sized to the same requirements with the calculator. Status is the paper result at TRL 3 from LVH-CAL-001; every requirement is verified by test at TRL 4.

Table 1. Requirements.

ID	Requirement	Target	Verification (TRL 4)	TRL 3 status
R1	Swamped canoe stays afloat with its design load	Design net load 118 kg (crew holding on, motor, gear, catch) carried for at least 1 h	Swamp test in sheltered water, based on the USCG level flotation method	Met on paper: carries 2.2 times the design load at 50 mm freeboard
R2	Freeboard when swamped	At least 50 mm at the lowest point of the gunwale, design load, fresh water	Measured during swamp test	Met on paper: 80 mm (105 mm amidships)
R3	Floats level when swamped	Heel and trim each within 10 deg, crew in the water not holding on, and with crew holding on both sides	Inclinometer reading during swamp test	Met on paper: 0.0 deg heel, 0.17 deg trim; 0.45 deg trim with crew holding on
R4	Crew can bail the swamped canoe	Two crew bail to 200 mm freeboard in 15 min or less, calm water	Timed bail-out trial	Met on paper (estimate): about 6 min
R5	Volume taken from the working space	10 % or less of the inside volume	Measured on the fitted canoe	Met: 7.6 %
R6	Material water uptake	Foam gains less than 5 % of its lift in mass after 30 days immersed	Immersion and weighing of samples	Met on supplier data (under 1 % by volume); to confirm by test
R7	Fixing strength	Each fixing holds 2 times its share of the full lift without pull-out	CalRig proof-load of a batten, screws and strap in a test panel	Met on paper: screw factor 5.5, D-ring 14.6, webbing 73
R8	Fitting	A local builder fits the kit in one working day with hand tools, modules made beforehand	Timed fitting with the partner builder	Met on paper (estimate): about 6.3 h
R9	Durability	No cracking or loss of lift after 12 months of normal use	Field inspection of pilot canoes	Cannot be shown on paper
R10	Cost	Parts for kits on two canoes against the value-engineering target of USD 1,000	Costed bill of materials	USD 1,350: over the value-engineering target by USD 350
R11	Reserve lift	At least 1.5 times the design net load carried at 50 mm freeboard	Swamp test with added weights	Met on paper: 2.2
R12	Damage tolerance	Still floats with the largest module lost and 5 % water uptake	Swamp test with one module removed	Met on paper: 9 mm freeboard, 1.5 deg heel
R13	Crew on one side	Heel within 10 deg with all four crew holding one grab line	Swamp test	Met on paper: 4.0 deg (low-side freeboard 31 mm)

■ Assumptions

- Reference canoe: planked, flat bottom 0.8 m wide, sides flared 31.6 deg, frames 50 x 60 mm at 600 mm, thwarts at 1.2, 3.0 and 4.8 m from the transom.
- Wet hull timber is counted as exactly as dense as water (no net lift from the hull).

- A crew member holding on with head and shoulders out puts 20 % of a 75 kg body on the canoe.
- Closed-cell polyethylene foam of about 30 kg/m³ with under 1 % water uptake by volume is available to the first partner.
- Owners accept 7.6 % of the inside volume, high along the sides, in exchange for flotation.
- The USCG level flotation method is adapted as a design benchmark only; the kit is not certified to it.

Safety: These requirements describe flotation for the boat, not for people. Life jackets remain the first line of defence; the swamp tests behind R1 to R4 and R11 to R13 are run in sheltered, shallow water with a safety boat and every person in a life jacket.